

Supplementary Data

Downstream Drug Response Prediction Reveals Biological Utility of Perturbation Response Models

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This Supplementary Data accompanies the main manuscript. It documents split designs, implementation details for the perturbation response models, reconstruction accuracy of predicted profiles, drug-response prediction from those profiles (stratified as in the main figures), and the corresponding statistical comparisons.

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1. Split design used for generalization evaluation

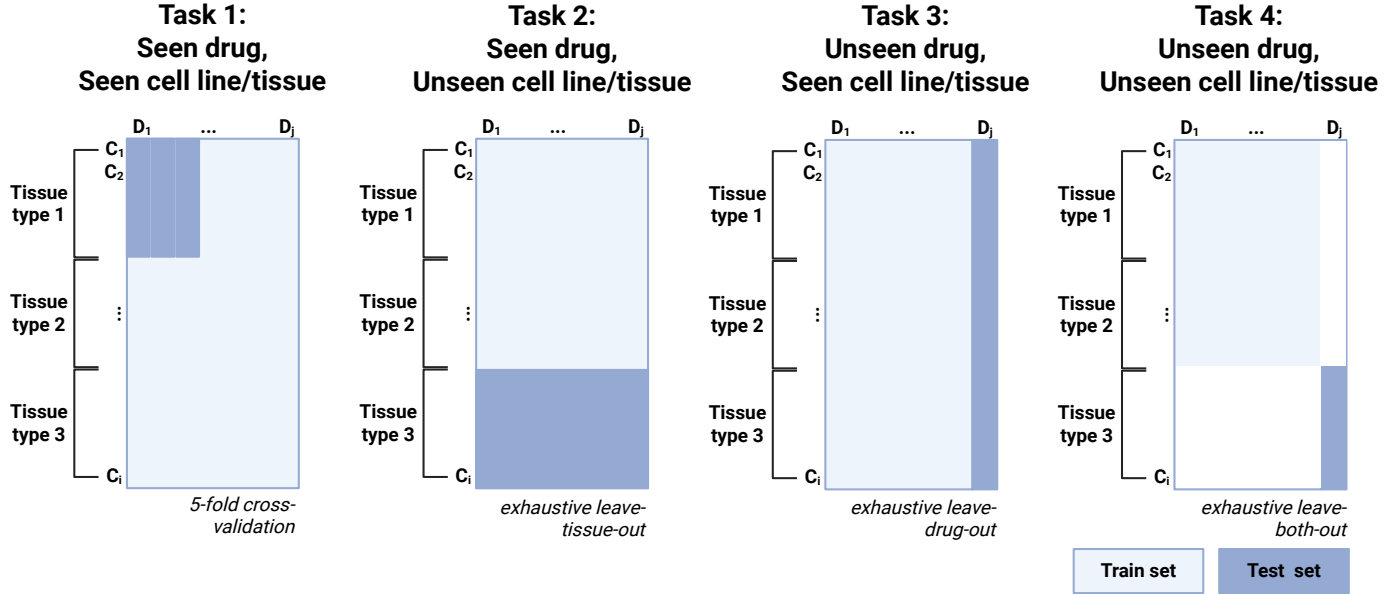


Figure 1: Split design used for generalization evaluation. For T2 to T4, we used exhaustive leave-one-out validation, holding out in turn each context (T2), each drug (T3), and both (T4). In these scenarios, the held-out unit is precisely what the model must generalize to. T1 asks whether responses can be interpolated across observed contexts and drugs, and holding out a single (cell line, drug) pair would leave the model trained on almost the entire dataset. We therefore used five-fold cross-validation across all five random folds over pairs for T1. Every pair appears exactly once in a held-out fold, resulting in independent predictions across the full dataset.

2. Methods description

Drug-response ElasticNet

Continuous sensitivity was predicted with scikit-learn ElasticNet after StandardScaler feature standardization. Hyperparameters were selected by inner 5-fold GridSearchCV on each outer training fold.

Table 1: ElasticNet drug-response predictor settings.

Setting	Value
Outer CV folds	5
Inner CV folds	5
Feature scaling	StandardScaler
alpha grid	{0.1, 1.0, 10.0}
l1_ratio grid	{0.0, 0.1, 0.5, 1.0}
max_iter	1000 (split-CV / response prediction); 2000 (measured-task CV)
Top variance genes	1000 (selected from pre-treatment only expression)
Morgan fingerprint	ECFP, radius 2, 2048 bits

CPA and CPA End-to-End

CPA used Gaussian reconstruction on log-normalized expression.

Table 2: CPA, chemCPA, and CPA End-to-End hyperparameters (SciPlex3 and McFarland).

Setting	Value
Latent dimension	128
Encoder / decoder	4 layers $\times$ 256 units
Batch / layer norm	batch norm on; layer norm off
Dropout	0.0
Doser	logsigm
Reconstruction loss	Gaussian
Adversary	4 layers $\times$ 64 units; reg_adv=20; pen_adv=5
AE pretraining epochs	25
Sensitivity head (end-to-end)	2 layers $\times$ 256 units
Learning rates (AE / adversary / doser / head)	0.003
Weight decay	4×10^{-7}
LR step size	45
Gradient clipping	value 1.0
Batch size	512
Max epochs	120
Training seed	0

chemCPA

chemCPA used the same architecture and optimizer settings as CPA (latent 128; AE depth 4 $\times$ width 256; adversary depth 4 $\times$ width 64; 120 epochs; learning rate 0.003; seed 0; batch size 512). Drug inputs were RDKit2D descriptors from SMILES. No LINCS pretraining was used (load_pretrained=False). Early-stopping patience was 15.

PRnet

Table 3: PRnet hyperparameters (SciPlex3 and McFarland CV).

Setting	Value
Latent dimension (z)	64
Hidden / adaptor layers	[128] / [128]
Combination dimension	64
Drug fingerprint	FCFP4 (Morgan radius 2, feature invariants), 1024 bits, scaled by $\log_{10}(\text{dose} + 1)$
Dropout	0.05
Loss	GUSS
Batch size	512
Epochs	100
Learning rate	1×10^{-3}
Weight decay	1×10^{-8}
LR scheduler	factor 0.5, patience 5
Within-fold validation fraction	0.1
Training seed	2024

GEARS

GEARS models used the published GEARS architecture with drug-target genes as perturbation labels.

Table 4: GEARS hyperparameters.

Setting	Value
Hidden size	64
Batch size / test batch size	32 / 128; 32 (end-to-end)
Epochs	20
Learning rate	1×10^{-3}
Optimizer weight decay (end-to-end)	5×10^{-4}
Training / fold seed	1

scFoundation

scFoundation was used as a frozen single-cell foundation encoder (`maeautobin`) inside a GEARS-style decoder; the foundation weights were not fine-tuned (`finetune_method=frozen`).

Table 5: scFoundation hyperparameters.

Setting	Value
Foundation model type	<code>maeautobin</code>
Binning	<code>autobin_resolution_append</code>
Hidden size	512
Epochs	20
Learning rate	2×10^{-4}
Train gene-set size	0.8
Mode / high-res	v1 / 0
Training seed	42

Random seeds

Table 6: Random seeds used across the pipeline.

Component	Seed
(Cell line, drug) CV split assignment	42
CPA / chemCPA / CPA End-to-End training	0
PRnet training	2024
GEARS / GEARS End-to-End training	1
scFoundation (SciPlex3)	1
Measured-task CV fallback fold RNG	1
Legacy shuffled KFold / StratifiedKFold helpers	42
GSEA permutations	42 (1,000 permutations)

Software and package versions

Analyses were run in Python conda environments. Key pinned versions are listed below; complete environment specifications are provided with the code.

Table 7: Principal software versions.

Component	Key versions
Drug-response pipeline	Python 3.10; scanpy; scikit-learn; anndata; PyTorch; cell-gears
CPA / CPA End-to-End	Python 3.10; PyTorch 2.0.1; CUDA 11.8; scvi-tools 0.20.3; anndata 0.9.2; scanpy 1.9.8; Lightning 2.2.x; NumPy 1.23.5
chemCPA	Python 3.9; PyTorch 1.12.1; CUDA toolkit 11.3; RDKit 2021.09.2; Hydra; scanpy
GEARS	GEARS 0.1.2; NumPy 1.26.4; pandas 2.2.2; scikit-learn 1.5.1; scanpy 1.10.2; SciPy 1.14.1; PyTorch; torch-geometric
PRnet	NumPy; pandas; scikit-learn; PyTorch; scanpy; RDKit; anndata
scFoundation	PyTorch; scanpy; frozen scFoundation checkpoint

3. Drug response prediction with measured profiles

Measured-profile ElasticNet: Pearson / Spearman / RMSE per stratification panel (mean $\pm$ s.d. across groups; paired Δ [95% CI] and BH-FDR q from two-sided paired t -tests vs Pre+Morgan within each panel across tasks $\times$ modalities; ** $q < 0.01$, * $q < 0.05$). Each panel's three metrics share one page.

Table 8: SciPlex3 within cell line: Pearson r (mean $\pm$ s.d.; paired Δ [95% CI], BH-FDR q vs Pre+Morgan within panel across tasks $\times$ modalities; ** $q < 0.01$, * $q < 0.05$).

Task	Modality	n	Mean $\pm$ s.d.	Δ vs Pre+Morgan [95% CI]	q
T1	Pre	3	0.58 $\pm$ 0.01	—	—
T1	Post	3	0.74 $\pm$ 0.04	+0.16 [0.05, 0.28]	0.0707
T1	LFC	3	0.77 $\pm$ 0.06	+0.19 [0.02, 0.37]	0.0812
T2	Pre	3	0.59 $\pm$ 0.02	—	—
T2	Post	3	0.18 $\pm$ 0.38	-0.42 [-1.37, 0.54]	0.324
T2	LFC	3	0.23 $\pm$ 0.38	-0.36 [-1.31, 0.59]	0.326
T3	Pre	3	0.29 $\pm$ 0.02	—	—
T3	Post	3	0.69 $\pm$ 0.06	+0.40 [0.25, 0.55]	0.0291*
T3	LFC	3	0.75 $\pm$ 0.07	+0.46 [0.30, 0.63]	0.0291*
T4	Pre	3	0.21 $\pm$ 0.17	—	—
T4	Post	3	0.05 $\pm$ 0.33	-0.16 [-1.35, 1.03]	0.708
T4	LFC	3	0.13 $\pm$ 0.37	-0.09 [-1.40, 1.22]	0.802

Table 9: SciPlex3 within cell line: Spearman ρ (mean $\pm$ s.d.; paired Δ [95% CI], BH-FDR q vs Pre+Morgan within panel across tasks $\times$ modalities; ** $q < 0.01$, * $q < 0.05$).

Task	Modality	n	Mean $\pm$ s.d.	Δ vs Pre+Morgan [95% CI]	q
T1	Pre	3	0.52 $\pm$ 0.02	—	—
T1	Post	3	0.64 $\pm$ 0.09	+0.12 [-0.14, 0.37]	0.25
T1	LFC	3	0.67 $\pm$ 0.08	+0.15 [-0.09, 0.39]	0.25
T2	Pre	3	0.56 $\pm$ 0.03	—	—
T2	Post	3	0.22 $\pm$ 0.28	-0.35 [-1.06, 0.37]	0.25
T2	LFC	3	0.31 $\pm$ 0.21	-0.25 [-0.80, 0.30]	0.25
T3	Pre	3	0.27 $\pm$ 0.07	—	—
T3	Post	3	0.61 $\pm$ 0.07	+0.34 [0.07, 0.61]	0.131
T3	LFC	3	0.67 $\pm$ 0.03	+0.40 [0.20, 0.60]	0.104
T4	Pre	3	0.24 $\pm$ 0.08	—	—
T4	Post	3	0.09 $\pm$ 0.23	-0.15 [-0.88, 0.58]	0.531
T4	LFC	3	0.19 $\pm$ 0.20	-0.05 [-0.71, 0.60]	0.77

Table 10: SciPlex3 within cell line: RMSE (mean $\pm$ s.d.; paired Δ [95% CI], BH-FDR q vs Pre+Morgan within panel across tasks $\times$ modalities; ** $q < 0.01$, * $q < 0.05$).

Task	Modality	n	Mean $\pm$ s.d.	Δ vs Pre+Morgan [95% CI]	q
T1	Pre	3	0.184 $\pm$ 0.046	—	—
T1	Post	3	0.149 $\pm$ 0.019	-0.035 [-0.102, 0.032]	0.31
T1	LFC	3	0.138 $\pm$ 0.013	-0.046 [-0.132, 0.040]	0.31
T2	Pre	3	0.538 $\pm$ 0.274	—	—
T2	Post	3	1.790 $\pm$ 1.207	+1.252 [-1.930, 4.433]	0.313
T2	LFC	3	0.347 $\pm$ 0.166	-0.191 [-0.842, 0.460]	0.335
T3	Pre	3	0.213 $\pm$ 0.045	—	—
T3	Post	3	0.158 $\pm$ 0.023	-0.055 [-0.113, 0.003]	0.232
T3	LFC	3	0.143 $\pm$ 0.018	-0.070 [-0.145, 0.006]	0.232
T4	Pre	3	0.550 $\pm$ 0.266	—	—
T4	Post	3	1.773 $\pm$ 1.195	+1.223 [-1.904, 4.351]	0.313
T4	LFC	3	0.358 $\pm$ 0.162	-0.192 [-0.834, 0.450]	0.335

Table 11: McFarland within tissue: Pearson r (mean $\pm$ s.d.; paired Δ [95% CI], BH-FDR q vs Pre+Morgan within panel across tasks $\times$ modalities; ** $q < 0.01$, * $q < 0.05$).

Task	Modality	n	Mean $\pm$ s.d.	Δ vs Pre+Morgan [95% CI]	q
T1	Pre	20	0.65 $\pm$ 0.14	—	—
T1	Post	20	0.68 $\pm$ 0.13	+0.04 [0.00, 0.07]	0.0482*
T1	LFC	20	0.73 $\pm$ 0.10	+0.08 [0.04, 0.13]	0.00134**
T2	Pre	20	0.65 $\pm$ 0.14	—	—
T2	Post	20	0.68 $\pm$ 0.13	+0.03 [-0.00, 0.07]	0.0643
T2	LFC	20	0.70 $\pm$ 0.13	+0.05 [-0.01, 0.11]	0.088
T3	Pre	20	-0.36 $\pm$ 0.19	—	—
T3	Post	20	-0.08 $\pm$ 0.27	+0.28 [0.15, 0.40]	3.5e-04**
T3	LFC	20	0.30 $\pm$ 0.22	+0.66 [0.50, 0.82]	2.2e-07**
T4	Pre	20	-0.38 $\pm$ 0.15	—	—
T4	Post	20	-0.04 $\pm$ 0.25	+0.34 [0.22, 0.46]	3.8e-05**
T4	LFC	20	0.28 $\pm$ 0.21	+0.66 [0.53, 0.80]	2.9e-08**

Table 12: McFarland within tissue: Spearman ρ (mean $\pm$ s.d.; paired Δ [95% CI], BH-FDR q vs Pre+Morgan within panel across tasks $\times$ modalities; ** $q < 0.01$, * $q < 0.05$).

Task	Modality	n	Mean $\pm$ s.d.	Δ vs Pre+Morgan [95% CI]	q
T1	Pre	20	0.54 $\pm$ 0.15	—	—
T1	Post	20	0.56 $\pm$ 0.20	+0.02 [-0.06, 0.09]	0.625
T1	LFC	20	0.63 $\pm$ 0.13	+0.09 [0.04, 0.13]	0.00215**
T2	Pre	20	0.55 $\pm$ 0.15	—	—
T2	Post	20	0.57 $\pm$ 0.15	+0.02 [-0.03, 0.08]	0.487
T2	LFC	20	0.60 $\pm$ 0.14	+0.06 [-0.00, 0.11]	0.0706
T3	Pre	20	-0.28 $\pm$ 0.22	—	—
T3	Post	20	-0.05 $\pm$ 0.25	+0.23 [0.09, 0.37]	0.00513**
T3	LFC	20	0.32 $\pm$ 0.18	+0.60 [0.44, 0.76]	9.1e-07**
T4	Pre	20	-0.32 $\pm$ 0.20	—	—
T4	Post	20	-0.03 $\pm$ 0.24	+0.30 [0.14, 0.46]	0.00192**
T4	LFC	20	0.31 $\pm$ 0.17	+0.63 [0.49, 0.78]	2.0e-07**

Table 13: McFarland within tissue: RMSE (mean $\pm$ s.d.; paired Δ [95% CI], BH-FDR q vs Pre+Morgan within panel across tasks $\times$ modalities; ** $q < 0.01$, * $q < 0.05$).

Task	Modality	n	Mean $\pm$ s.d.	Δ vs Pre+Morgan [95% CI]	q
T1	Pre	20	0.153 $\pm$ 0.037	—	—
T1	Post	20	0.145 $\pm$ 0.033	-0.007 [-0.012, -0.003]	0.0053**
T1	LFC	20	0.136 $\pm$ 0.030	-0.016 [-0.023, -0.010]	7.5e-05**
T2	Pre	20	0.156 $\pm$ 0.038	—	—
T2	Post	20	0.151 $\pm$ 0.036	-0.005 [-0.010, 0.000]	0.0537
T2	LFC	20	0.144 $\pm$ 0.033	-0.011 [-0.019, -0.004]	0.0059**
T3	Pre	20	0.220 $\pm$ 0.033	—	—
T3	Post	20	0.207 $\pm$ 0.034	-0.013 [-0.018, -0.008]	1.2e-04**
T3	LFC	20	0.192 $\pm$ 0.033	-0.028 [-0.038, -0.018]	3.0e-05**
T4	Pre	20	0.221 $\pm$ 0.033	—	—
T4	Post	20	0.210 $\pm$ 0.035	-0.011 [-0.017, -0.006]	6.3e-04**
T4	LFC	20	0.195 $\pm$ 0.035	-0.026 [-0.035, -0.017]	3.0e-05**

Table 14: McFarland within drug: Pearson r (mean $\pm$ s.d.; paired Δ [95% CI], BH-FDR q vs Pre+Morgan within panel across tasks $\times$ modalities; ** $q < 0.01$, * $q < 0.05$).

Task	Modality	n	Mean $\pm$ s.d.	Δ vs Pre+Morgan [95% CI]	q
T1	Pre	11	-0.01 $\pm$ 0.29	—	—
T1	Post	11	0.29 $\pm$ 0.27	+0.30 [0.06, 0.53]	0.0198*
T1	LFC	11	0.43 $\pm$ 0.29	+0.43 [0.15, 0.72]	0.0105*
T2	Pre	11	-0.18 $\pm$ 0.17	—	—
T2	Post	11	0.19 $\pm$ 0.26	+0.37 [0.14, 0.60]	0.0105*
T2	LFC	11	0.36 $\pm$ 0.27	+0.53 [0.29, 0.78]	0.00578**
T3	Pre	11	-0.02 $\pm$ 0.23	—	—
T3	Post	11	0.27 $\pm$ 0.27	+0.29 [0.06, 0.52]	0.0198*
T3	LFC	11	0.43 $\pm$ 0.33	+0.44 [0.16, 0.73]	0.0105*
T4	Pre	11	-0.07 $\pm$ 0.13	—	—
T4	Post	11	0.20 $\pm$ 0.27	+0.27 [0.03, 0.51]	0.0308*
T4	LFC	11	0.37 $\pm$ 0.28	+0.44 [0.19, 0.68]	0.01*

Table 15: McFarland within drug: Spearman ρ (mean $\pm$ s.d.; paired Δ [95% CI], BH-FDR q vs Pre+Morgan within panel across tasks $\times$ modalities; ** $q < 0.01$, * $q < 0.05$).

Task	Modality	n	Mean $\pm$ s.d.	Δ vs Pre+Morgan [95% CI]	q
T1	Pre	11	-0.00 $\pm$ 0.27	—	—
T1	Post	11	0.24 $\pm$ 0.22	+0.24 [0.03, 0.45]	0.028*
T1	LFC	11	0.35 $\pm$ 0.19	+0.35 [0.12, 0.58]	0.0109*
T2	Pre	11	-0.15 $\pm$ 0.19	—	—
T2	Post	11	0.19 $\pm$ 0.20	+0.34 [0.12, 0.56]	0.0109*
T2	LFC	11	0.29 $\pm$ 0.18	+0.44 [0.26, 0.63]	0.00205**
T3	Pre	11	-0.03 $\pm$ 0.21	—	—
T3	Post	11	0.21 $\pm$ 0.21	+0.24 [0.03, 0.44]	0.028*
T3	LFC	11	0.35 $\pm$ 0.22	+0.38 [0.16, 0.60]	0.00883**
T4	Pre	11	-0.04 $\pm$ 0.11	—	—
T4	Post	11	0.18 $\pm$ 0.23	+0.22 [0.03, 0.41]	0.028*
T4	LFC	11	0.31 $\pm$ 0.18	+0.35 [0.19, 0.50]	0.00205**

Table 16: McFarland within drug: RMSE (mean $\pm$ s.d.; paired Δ [95% CI], BH-FDR q vs Pre+Morgan within panel across tasks $\times$ modalities; ** $q < 0.01$, * $q < 0.05$).

Task	Modality	n	Mean $\pm$ s.d.	Δ vs Pre+Morgan [95% CI]	q
T1	Pre	11	0.146 $\pm$ 0.031	—	—
T1	Post	11	0.135 $\pm$ 0.033	-0.011 [-0.020, -0.001]	0.0858
T1	LFC	11	0.125 $\pm$ 0.035	-0.021 [-0.035, -0.006]	0.0536
T2	Pre	11	0.148 $\pm$ 0.034	—	—
T2	Post	11	0.139 $\pm$ 0.038	-0.009 [-0.019, 0.001]	0.135
T2	LFC	11	0.131 $\pm$ 0.037	-0.017 [-0.030, -0.004]	0.0536
T3	Pre	11	0.211 $\pm$ 0.081	—	—
T3	Post	11	0.196 $\pm$ 0.079	-0.015 [-0.046, 0.015]	0.292
T3	LFC	11	0.180 $\pm$ 0.082	-0.031 [-0.079, 0.017]	0.236
T4	Pre	11	0.213 $\pm$ 0.080	—	—
T4	Post	11	0.198 $\pm$ 0.081	-0.015 [-0.042, 0.012]	0.27
T4	LFC	11	0.185 $\pm$ 0.082	-0.028 [-0.070, 0.013]	0.236

4. Reconstruction accuracy metrics

Mean $\pm$ s.d. of reconstruction metrics across cell $\times$ condition pairs for SciPlex3 and McFarland. Median Δ versus Average effect / No effect with BH-FDR (paired Wilcoxon, else Mann–Whitney) is included for each PRM.

Table 17: Reconstruction: MSE (all genes) (mean $\pm$ s.d.; median Δ vs Avg / Noeff; BH-FDR; ** FDR < 0.01, * FDR < 0.05).

Dataset	Model	Mean $\pm$ s.d.	Median Δ vs Avg	FDR	Median Δ vs Noeff	FDR
SciPlex3	CPA	8.30e-04 $\pm$ 4.25e-04	+6.69e-04	1.49e-18	+6.58e-04	1.28e-17
SciPlex3	chemCPA	7.53e-04 $\pm$ 4.73e-04	+4.99e-04	3.72e-18	+5.04e-04	2.35e-17
SciPlex3	PRnet	1.11e-03 $\pm$ 6.58e-04	+6.12e-04	4.52e-19	+5.91e-04	8.90e-19
SciPlex3	GEARS	5.56e-04 $\pm$ 9.43e-04	+8.84e-05	1.03e-36	+8.76e-05	2.54e-35
SciPlex3	scFoundation	1.21e-02 $\pm$ 1.37e-02	+2.39e-03	5.98e-39	+2.37e-03	5.98e-39
SciPlex3	Average effect	4.10e-04 $\pm$ 7.55e-04	—	—	—	—
SciPlex3	No effect	4.46e-04 $\pm$ 8.37e-04	—	—	—	—
McFarland	CPA	3.36e-03 $\pm$ 1.68e-03	-1.32e-03	1.52e-76**	-5.07e-03	1.87e-155**
McFarland	chemCPA	4.73e-03 $\pm$ 1.86e-03	-3.86e-06	1.07e-03**	-3.75e-03	2.26e-125**
McFarland	PRnet	4.68e-03 $\pm$ 1.79e-03	+2.60e-06	4.59e-04	-3.71e-03	1.06e-118**
McFarland	GEARS	1.27e-02 $\pm$ 7.89e-03	+5.26e-03	4.37e-41	+3.82e-04	5.18e-04
McFarland	scFoundation	2.13e-02 $\pm$ 9.03e-03	+1.31e-02	3.58e-61	+8.29e-03	1.48e-58
McFarland	Average effect	5.50e-03 $\pm$ 3.84e-03	—	—	—	—
McFarland	No effect	1.03e-02 $\pm$ 7.64e-03	—	—	—	—

Table 18: Reconstruction: Pearson r (all genes) (mean $\pm$ s.d.; median Δ vs Avg / Noeff; BH-FDR; ** FDR < 0.01, * FDR < 0.05).

Dataset	Model	Mean $\pm$ s.d.	Median Δ vs Avg	FDR	Median Δ vs Noeff	FDR
SciPlex3	CPA	0.98 $\pm$ 0.02	-0.01	1.26e-14	-0.01	9.81e-11
SciPlex3	chemCPA	0.98 $\pm$ 0.02	-0.00	5.90e-14	-0.00	3.63e-08
SciPlex3	PRnet	0.97 $\pm$ 0.02	-0.02	6.36e-16	-0.02	8.13e-16
SciPlex3	GEARS	0.97 $\pm$ 0.04	-0.01	8.75e-37	-0.01	6.09e-35
SciPlex3	scFoundation	0.77 $\pm$ 0.18	-0.09	5.98e-39	-0.09	5.98e-39
SciPlex3	Average effect	0.98 $\pm$ 0.03	—	—	—	—
SciPlex3	No effect	0.98 $\pm$ 0.04	—	—	—	—
McFarland	CPA	0.97 $\pm$ 0.02	-0.02	2.74e-140	-0.01	1.96e-52
McFarland	chemCPA	0.95 $\pm$ 0.02	-0.03	5.21e-157	-0.02	1.56e-136
McFarland	PRnet	0.95 $\pm$ 0.02	-0.03	2.61e-156	-0.02	8.57e-124
McFarland	GEARS	0.96 $\pm$ 0.02	-0.01	1.56e-39	-0.00	4.73e-04
McFarland	scFoundation	0.94 $\pm$ 0.03	-0.03	1.79e-61	-0.02	2.91e-58
McFarland	Average effect	0.98 $\pm$ 0.01	—	—	—	—
McFarland	No effect	0.97 $\pm$ 0.02	—	—	—	—

Table 19: Reconstruction: MSE (top20 DE) (mean $\pm$ s.d.; median Δ vs Avg / Noeff; BH-FDR; ** FDR < 0.01, * FDR < 0.05).

Dataset	Model	Mean $\pm$ s.d.	Median Δ vs Avg	FDR	Median Δ vs Noeff	FDR
SciPlex3	CPA	1.16e-02 $\pm$ 3.13e-02	+1.48e-04	1.08e-02	+1.21e-04	1.06e-01
SciPlex3	chemCPA	1.38e-02 $\pm$ 3.89e-02	+1.02e-04	4.54e-04	+9.27e-05	6.52e-03
SciPlex3	PRnet	1.38e-02 $\pm$ 3.76e-02	-1.57e-04	5.65e-06**	-1.86e-04	4.68e-07**
SciPlex3	GEARS	2.95e-02 $\pm$ 1.04e-01	+8.12e-05	8.44e-12	+1.60e-05	1.81e-01
SciPlex3	scFoundation	4.15e-02 $\pm$ 9.29e-02	+2.69e-03	1.39e-25	+2.36e-03	7.95e-21
SciPlex3	Average effect	2.94e-02 $\pm$ 1.02e-01	—	—	—	—
SciPlex3	No effect	3.37e-02 $\pm$ 1.13e-01	—	—	—	—
McFarland	CPA	1.03e-01 $\pm$ 1.05e-01	-1.70e-01	5.66e-150**	-5.50e-01	7.13e-163**
McFarland	chemCPA	1.33e-01 $\pm$ 1.13e-01	-1.38e-01	2.85e-139**	-5.14e-01	2.27e-161**
McFarland	PRnet	1.01e-01 $\pm$ 1.01e-01	-1.71e-01	4.08e-144**	-5.36e-01	2.27e-161**
McFarland	GEARS	1.04e+00 $\pm$ 1.02e+00	+3.23e-01	9.46e-39	-6.42e-02	7.03e-15**
McFarland	scFoundation	8.12e-01 $\pm$ 8.58e-01	+1.79e-01	6.80e-44	-2.92e-01	8.46e-37**
McFarland	Average effect	4.12e-01 $\pm$ 5.32e-01	—	—	—	—
McFarland	No effect	8.47e-01 $\pm$ 8.04e-01	—	—	—	—

Table 20: Reconstruction: Pearson r (top20 DE) (mean $\pm$ s.d.; median Δ vs Avg / Noeff; BH-FDR; ** FDR < 0.01, * FDR < 0.05).

Dataset	Model	Mean $\pm$ s.d.	Median Δ vs Avg	FDR	Median Δ vs Noeff	FDR
SciPlex3	CPA	0.57 $\pm$ 0.35	-0.10	3.52e-14	-0.16	7.87e-14
SciPlex3	chemCPA	0.73 $\pm$ 0.28	-0.01	1.08e-05	-0.03	2.64e-09
SciPlex3	PRnet	0.39 $\pm$ 0.47	-0.33	3.52e-14	-0.29	8.54e-14
SciPlex3	GEARS	0.64 $\pm$ 0.35	-0.03	1.92e-23	-0.03	4.43e-18
SciPlex3	scFoundation	0.40 $\pm$ 0.45	-0.22	5.40e-35	-0.25	1.36e-34
SciPlex3	Average effect	0.77 $\pm$ 0.28	—	—	—	—
SciPlex3	No effect	0.80 $\pm$ 0.27	—	—	—	—
McFarland	CPA	0.60 $\pm$ 0.29	-0.18	1.48e-97	-0.06	3.73e-12
McFarland	chemCPA	0.46 $\pm$ 0.34	-0.31	5.41e-138	-0.18	2.24e-73
McFarland	PRnet	0.61 $\pm$ 0.29	-0.17	1.37e-86	-0.07	2.50e-10
McFarland	GEARS	0.64 $\pm$ 0.30	-0.10	6.20e-36	+0.02	1.39e-13**
McFarland	scFoundation	0.70 $\pm$ 0.26	-0.06	4.08e-45	+0.09	1.63e-27**
McFarland	Average effect	0.82 $\pm$ 0.21	—	—	—	—
McFarland	No effect	0.66 $\pm$ 0.30	—	—	—	—

Table 21: Reconstruction: Systema Pearson r (per cell line) (mean $\pm$ s.d.; median Δ vs Avg / Noeff; BH-FDR; ** FDR < 0.01, * FDR < 0.05).

Dataset	Model	Mean $\pm$ s.d.	Median Δ vs Avg	FDR	Median Δ vs Noeff	FDR
SciPlex3	CPA	0.09 $\pm$ 0.19	+0.08	4.21e-10**	-0.10	5.98e-03
SciPlex3	chemCPA	0.03 $\pm$ 0.19	+0.04	4.33e-05**	-0.13	3.26e-32
SciPlex3	PRnet	0.04 $\pm$ 0.17	+0.04	1.08e-03**	-0.14	6.03e-06
SciPlex3	GEARS	0.05 $\pm$ 0.11	+0.05	7.65e-14**	-0.15	3.09e-08
SciPlex3	scFoundation	0.01 $\pm$ 0.06	+0.00	3.41e-01	-0.20	8.04e-13
SciPlex3	Average effect	0.00 $\pm$ 0.03	—	—	—	—
SciPlex3	No effect	0.13 $\pm$ 0.29	—	—	—	—
McFarland	CPA	0.22 $\pm$ 0.21	+0.19	1.22e-129**	+0.13	7.10e-40**
McFarland	chemCPA	0.05 $\pm$ 0.17	+0.05	2.52e-72**	+0.01	9.54e-02
McFarland	PRnet	0.17 $\pm$ 0.19	+0.16	2.26e-105**	+0.09	1.36e-18**
McFarland	GEARS	0.01 $\pm$ 0.15	-0.04	1.76e-01	+0.07	2.53e-01
McFarland	scFoundation	-0.06 $\pm$ 0.09	-0.06	3.77e-24	+0.01	1.18e-01
McFarland	Average effect	-0.00 $\pm$ 0.05	—	—	—	—
McFarland	No effect	0.04 $\pm$ 0.35	—	—	—	—

5. Drug response prediction with predicted profiles

Mean $\pm$ s.d. of Pearson r , Spearman ρ , and RMSE for drug-aware modalities, stratified by drug or by tissue/cell line (McFarland tissue, SciPlex3 cell line), matching the stratified figure panels. Paired Δ [95% CI] versus Pre+Morgan, Average effect, and No effect are included in each table. Primary comparisons (CPA vs Pre+Morgan and vs Average effect, Pearson) report BH-FDR q within each stratum; all other p -values (other PRMs, No effect, Spearman, RMSE) are uncorrected and exploratory.

Per drug

McFarland only (SciPlex3 omitted: too few cell lines per drug for stable within-drug metrics).

Table 22: Within-drug (McFarland): Pearson r (mean $\pm$ s.d.; paired Δ [95% CI]. Primary: CPA vs Pre+Morgan / Average effect (Pearson) reports BH-FDR q within stratum; all other p are uncorrected exploratory. Sig: ** < 0.01 , * < 0.05).

Dataset	Model	Modality	Mean $\pm$ s.d.	Δ vs Pre [95% CI]	p/q	Δ vs Avg [95% CI]	p/q	Δ vs Noeff [95% CI]	p/q
McFarland	Average effect	LFC+Morgan	0.06 $\pm$ 0.27	+0.06 [-0.04, 0.17]	0.205	—	—	+0.27 [0.12, 0.42]	0.0024**
McFarland	Average effect	Post+Morgan	-0.03 $\pm$ 0.29	-0.02 [-0.05, 0.01]	0.103	—	—	-0.02 [-0.05, 0.01]	0.104
McFarland	CPA	LFC+Morgan	0.30 $\pm$ 0.24	+0.30 [0.06, 0.54]	0.0433*	+0.24 [0.04, 0.44]	0.0433*	+0.51 [0.28, 0.74]	6.5e-04**
McFarland	CPA	Post+Morgan	0.15 $\pm$ 0.25	+0.16 [-0.03, 0.34]	0.0845	+0.18 [0.01, 0.36]	0.0549	+0.16 [-0.03, 0.34]	0.0844
McFarland	CPA End-to-End	-	0.02 $\pm$ 0.14	+0.03 [-0.15, 0.21]	0.745	-0.04 [-0.18, 0.11]	0.589	+0.23 [0.10, 0.37]	0.00362**
McFarland	ElasticNet	LFC+Morgan	0.43 $\pm$ 0.29	+0.43 [0.15, 0.72]	0.00658**	+0.37 [0.14, 0.61]	0.0057**	+0.64 [0.37, 0.92]	4.0e-04**
McFarland	ElasticNet	Post+Morgan	0.29 $\pm$ 0.27	+0.30 [0.06, 0.53]	0.0173*	+0.32 [0.10, 0.54]	0.00933**	+0.30 [0.06, 0.53]	0.0173*
McFarland	ElasticNet	Pre+Morgan	-0.01 $\pm$ 0.29	—	—	—	—	—	—
McFarland	GEARS	LFC+Morgan	-0.14 $\pm$ 0.23	-0.15 [-0.38, 0.07]	0.158	-0.20 [-0.37, -0.03]	0.0249*	+0.07 [-0.15, 0.29]	0.474
McFarland	GEARS	Post+Morgan	-0.09 $\pm$ 0.23	-0.10 [-0.30, 0.10]	0.294	-0.07 [-0.28, 0.14]	0.457	-0.10 [-0.30, 0.10]	0.294
McFarland	No effect	LFC+Morgan	-0.21 $\pm$ 0.19	-0.21 [-0.32, -0.09]	0.00244**	-0.27 [-0.42, -0.12]	0.0024**	—	—
McFarland	No effect	Post+Morgan	-0.01 $\pm$ 0.29	-0.00 [-0.00, 0.00]	0.306	+0.02 [-0.01, 0.05]	0.104	—	—
McFarland	PRnet	LFC+Morgan	-0.01 $\pm$ 0.21	-0.00 [-0.07, 0.06]	0.879	-0.07 [-0.18, 0.05]	0.215	+0.20 [0.14, 0.27]	3.4e-05**
McFarland	PRnet	Post+Morgan	-0.00 $\pm$ 0.12	+0.00 [-0.18, 0.18]	0.998	+0.02 [-0.16, 0.20]	0.774	+0.00 [-0.18, 0.18]	0.998
McFarland	chemCPA	LFC+Morgan	0.01 $\pm$ 0.18	+0.02 [-0.06, 0.10]	0.619	-0.04 [-0.15, 0.06]	0.378	+0.23 [0.16, 0.29]	2.5e-05**
McFarland	chemCPA	Post+Morgan	-0.05 $\pm$ 0.28	-0.05 [-0.13, 0.04]	0.237	-0.02 [-0.11, 0.06]	0.547	-0.05 [-0.13, 0.04]	0.238
McFarland	scFoundation	LFC+Morgan	0.02 $\pm$ 0.24	+0.03 [-0.18, 0.24]	0.768	-0.03 [-0.21, 0.14]	0.669	+0.24 [0.03, 0.44]	0.0306*
McFarland	scFoundation	Post+Morgan	-0.03 $\pm$ 0.25	-0.02 [-0.21, 0.17]	0.79	+0.00 [-0.19, 0.19]	0.995	-0.02 [-0.21, 0.17]	0.79

Table 23: Within-drug (McFarland): Spearman ρ (mean $\pm$ s.d.; paired Δ [95% CI]. Primary: CPA vs Pre+Morgan / Average effect (Pearson) reports BH-FDR q within stratum; all other p are uncorrected exploratory. Sig: ** < 0.01 , * < 0.05).

Dataset	Model	Modality	Mean $\pm$ s.d.	Δ vs Pre [95% CI]	p/q	Δ vs Avg [95% CI]	p/q	Δ vs Noeff [95% CI]	p/q
McFarland	Average effect	LFC+Morgan	0.02 $\pm$ 0.20	+0.03 [-0.07, 0.13]	0.566	—	—	+0.22 [0.08, 0.35]	0.00465**
McFarland	Average effect	Post+Morgan	-0.03 $\pm$ 0.25	-0.02 [-0.06, 0.01]	0.212	—	—	-0.02 [-0.06, 0.01]	0.206
McFarland	CPA	LFC+Morgan	0.24 $\pm$ 0.22	+0.24 [0.03, 0.46]	0.0292*	+0.22 [0.06, 0.38]	0.0129*	+0.44 [0.21, 0.66]	0.00137**
McFarland	CPA	Post+Morgan	0.11 $\pm$ 0.24	+0.12 [-0.04, 0.28]	0.131	+0.14 [-0.00, 0.28]	0.0531	+0.12 [-0.04, 0.28]	0.132
McFarland	CPA End-to-End	-	0.03 $\pm$ 0.16	+0.04 [-0.16, 0.23]	0.668	+0.01 [-0.10, 0.12]	0.827	+0.23 [0.05, 0.41]	0.0188*
McFarland	ElasticNet	LFC+Morgan	0.35 $\pm$ 0.19	+0.35 [0.12, 0.58]	0.00627**	+0.33 [0.14, 0.51]	0.00289**	+0.54 [0.33, 0.75]	1.7e-04**
McFarland	ElasticNet	Post+Morgan	0.24 $\pm$ 0.22	+0.24 [0.03, 0.45]	0.028*	+0.26 [0.07, 0.46]	0.0126*	+0.24 [0.03, 0.45]	0.0281*
McFarland	ElasticNet	Pre+Morgan	-0.00 $\pm$ 0.27	—	—	—	—	—	—
McFarland	GEARS	LFC+Morgan	-0.17 $\pm$ 0.17	-0.17 [-0.35, 0.01]	0.0622	-0.18 [-0.30, -0.07]	0.00585**	+0.03 [-0.15, 0.22]	0.701
McFarland	GEARS	Post+Morgan	-0.06 $\pm$ 0.24	-0.07 [-0.27, 0.14]	0.49	-0.04 [-0.24, 0.17]	0.695	-0.07 [-0.27, 0.14]	0.49
McFarland	No effect	LFC+Morgan	-0.19 $\pm$ 0.18	-0.19 [-0.31, -0.07]	0.00518**	-0.22 [-0.35, -0.08]	0.00465**	—	—
McFarland	No effect	Post+Morgan	-0.00 $\pm$ 0.27	+0.00 [-0.00, 0.00]	0.157	+0.02 [-0.01, 0.06]	0.206	—	—
McFarland	PRnet	LFC+Morgan	-0.01 $\pm$ 0.19	-0.00 [-0.07, 0.07]	0.952	-0.03 [-0.12, 0.06]	0.505	+0.19 [0.11, 0.27]	3.7e-04**
McFarland	PRnet	Post+Morgan	0.01 $\pm$ 0.15	+0.02 [-0.14, 0.18]	0.822	+0.04 [-0.11, 0.18]	0.568	+0.02 [-0.14, 0.18]	0.824
McFarland	chemCPA	LFC+Morgan	0.02 $\pm$ 0.17	+0.02 [-0.06, 0.10]	0.531	-0.00 [-0.08, 0.08]	0.922	+0.21 [0.13, 0.29]	1.4e-04**
McFarland	chemCPA	Post+Morgan	-0.06 $\pm$ 0.26	-0.05 [-0.15, 0.04]	0.24	-0.03 [-0.12, 0.06]	0.442	-0.05 [-0.15, 0.04]	0.238
McFarland	scFoundation	LFC+Morgan	0.04 $\pm$ 0.29	+0.04 [-0.15, 0.23]	0.632	+0.02 [-0.11, 0.14]	0.79	+0.23 [0.00, 0.46]	0.0476*
McFarland	scFoundation	Post+Morgan	0.01 $\pm$ 0.27	+0.01 [-0.17, 0.19]	0.9	+0.03 [-0.14, 0.20]	0.68	+0.01 [-0.17, 0.19]	0.902

Table 24: Within-drug (McFarland): RMSE (mean $\pm$ s.d.; paired Δ [95% CI]. Primary: CPA vs Pre+Morgan / Average effect (Pearson) reports BH-FDR q within stratum; all other p are uncorrected exploratory. Sig: ** < 0.01 , * < 0.05).

Dataset	Model	Modality	Mean $\pm$ s.d.	Δ vs Pre [95% CI]	p/q	Δ vs Avg [95% CI]	p/q	Δ vs Noeff [95% CI]	p/q
McFarland	Average effect	LFC+Morgan	0.145 $\pm$ 0.031	-0.000 [-0.004, 0.004] +0.000	0.851	—	—	+0.002 [-0.004, 0.009] +0.000	0.44
McFarland	Average effect	Post+Morgan	0.146 $\pm$ 0.031	-0.000 [-0.000, 0.001] -0.009	0.325	—	—	-0.007 [-0.000, 0.001] -0.007	0.32
McFarland	CPA	LFC+Morgan	0.136 $\pm$ 0.032	[-0.018, -0.000] -0.003	0.0467*	[-0.017, -0.001] -0.004	0.0302*	[-0.016, 0.003] -0.003	0.168
McFarland	CPA	Post+Morgan	0.142 $\pm$ 0.031	[-0.010, 0.003] +0.020	0.278	[-0.010, 0.003] +0.021	0.211	[-0.010, 0.003] +0.023	0.279
McFarland	CPA End-to-End	-	0.166 $\pm$ 0.035	[0.012, 0.028] -0.021	1.6e-04**	[0.013, 0.028] -0.020	1.2e-04**	[0.012, 0.034] -0.018	0.00103**
McFarland	ElasticNet	LFC+Morgan	0.125 $\pm$ 0.035	[-0.035, -0.006] -0.011	0.00932**	[-0.033, -0.008] -0.011	0.00534**	[-0.033, -0.003] -0.011	0.0214*
McFarland	ElasticNet	Post+Morgan	0.135 $\pm$ 0.033	[-0.020, -0.001] —	0.0322*	[-0.020, -0.002] —	0.0222*	[-0.020, -0.001] —	0.0322*
McFarland	ElasticNet	Pre+Morgan	0.146 $\pm$ 0.031	—	—	—	—	—	—
McFarland	GEARS	LFC+Morgan	0.158 $\pm$ 0.034	+0.010 [0.001, 0.019] +0.006	0.0339*	+0.010 [0.003, 0.018] +0.005	0.0145*	+0.012 [0.003, 0.021] +0.006	0.0116*
McFarland	GEARS	Post+Morgan	0.153 $\pm$ 0.036	[-0.003, 0.015] -0.003	0.193	[-0.004, 0.015] -0.002	0.228	[-0.003, 0.015] —	0.193
McFarland	No effect	LFC+Morgan	0.143 $\pm$ 0.035	[-0.008, 0.003] -0.000	0.297	[-0.009, 0.004] -0.000	0.44	—	—
McFarland	No effect	Post+Morgan	0.146 $\pm$ 0.031	[-0.000, 0.000] +0.001	0.306	[-0.001, 0.000] +0.002	0.32	—	—
McFarland	PRnet	LFC+Morgan	0.147 $\pm$ 0.032	[-0.000, 0.003] +0.048	0.124	[-0.004, 0.007] +0.047	0.51	[-0.002, 0.010] +0.048	0.179
McFarland	PRnet	Post+Morgan	0.194 $\pm$ 0.070	[0.012, 0.083] -0.001	0.0136*	[0.012, 0.083] -0.000	0.0141*	[0.012, 0.083] +0.002	0.0135*
McFarland	chemCPA	LFC+Morgan	0.145 $\pm$ 0.032	[-0.003, 0.002] -0.000	0.496	[-0.005, 0.004] -0.000	0.863	[-0.003, 0.007] -0.000	0.352
McFarland	chemCPA	Post+Morgan	0.146 $\pm$ 0.033	[-0.002, 0.002] -0.002	0.932	[-0.003, 0.002] -0.002	0.696	[-0.002, 0.002] +0.001	0.934
McFarland	scFoundation	LFC+Morgan	0.144 $\pm$ 0.038	[-0.014, 0.010] +0.002	0.729	[-0.013, 0.010] +0.002	0.772	[-0.010, 0.012] +0.002	0.864
McFarland	scFoundation	Post+Morgan	0.148 $\pm$ 0.040	[-0.011, 0.015] —	0.708	[-0.011, 0.015] —	0.751	[-0.011, 0.015] —	0.708

Per tissue / cell line

SciPlex3 groups are cell lines; McFarland groups are tissues (folds pooled within each group).

SciPlex3 per cell line

Table 25: Within cell line (SciPlex3): Pearson r (mean $\pm$ s.d.; paired Δ [95% CI]. Primary: CPA vs Pre+Morgan / Average effect (Pearson) reports BH-FDR q within stratum; all other p are uncorrected exploratory. Sig: ** < 0.01 , * < 0.05).

Dataset	Model	Modality	Mean $\pm$ s.d.	Δ vs Pre [95% CI]	p/q	Δ vs Avg [95% CI]	p/q	Δ vs Noeff [95% CI]	p/q
SciPlex3	Average effect	LFC+Morgan	0.54 $\pm$ 0.01	+0.00 [-0.00, 0.00]	0.397	—	—	-0.04 [-0.07, -0.02]	0.0188*
SciPlex3	Average effect	Post+Morgan	0.54 $\pm$ 0.01	+0.00 [-0.00, 0.00]	0.823	—	—	+0.00 [-0.00, 0.00]	0.823
SciPlex3	CPA	LFC+Morgan	0.71 $\pm$ 0.06	+0.16 [0.00, 0.33]	0.0535	+0.16 [0.00, 0.33]	0.0535	+0.12 [-0.04, 0.28]	0.0832
SciPlex3	CPA	Post+Morgan	0.69 $\pm$ 0.06	+0.15 [-0.01, 0.30]	0.0535	+0.15 [-0.01, 0.30]	0.0535	+0.15 [-0.01, 0.30]	0.0535
SciPlex3	CPA End-to-End	-	0.40 $\pm$ 0.08	-0.14 [-0.37, 0.09]	0.116	-0.14 [-0.37, 0.09]	0.116	-0.18 [-0.39, 0.02]	0.0598
SciPlex3	ElasticNet	LFC+Morgan	0.77 $\pm$ 0.06	+0.23 [0.07, 0.39]	0.026*	+0.23 [0.07, 0.39]	0.026*	+0.19 [0.03, 0.35]	0.0378*
SciPlex3	ElasticNet	Post+Morgan	0.74 $\pm$ 0.04	+0.20 [0.10, 0.30]	0.0137*	+0.20 [0.10, 0.30]	0.0137*	+0.20 [0.10, 0.30]	0.0137*
SciPlex3	ElasticNet	Pre+Morgan	0.54 $\pm$ 0.01	—	—	—	—	—	—
SciPlex3	GEARS	LFC+Morgan	0.63 $\pm$ 0.03	+0.09 [0.05, 0.14]	0.0123*	+0.09 [0.05, 0.14]	0.0124*	+0.05 [-0.02, 0.12]	0.0848
SciPlex3	GEARS	Post+Morgan	0.64 $\pm$ 0.04	+0.09 [0.04, 0.14]	0.0165*	+0.09 [0.04, 0.14]	0.0165*	+0.09 [0.04, 0.14]	0.0165*
SciPlex3	No effect	LFC+Morgan	0.59 $\pm$ 0.01	+0.04 [0.02, 0.07]	0.0189*	+0.04 [0.02, 0.07]	0.0188*	—	—
SciPlex3	No effect	Post+Morgan	0.54 $\pm$ 0.01	+0.00 [0.00, 0.00]	—	-0.00 [-0.00, 0.00]	0.823	—	—
SciPlex3	PRnet	LFC+Morgan	0.57 $\pm$ 0.07	+0.03 [-0.16, 0.23]	0.566	+0.03 [-0.16, 0.23]	0.567	-0.01 [-0.20, 0.18]	0.819
SciPlex3	PRnet	Post+Morgan	0.61 $\pm$ 0.05	+0.07 [-0.07, 0.22]	0.172	+0.07 [-0.07, 0.22]	0.172	+0.07 [-0.07, 0.22]	0.172
SciPlex3	chemCPA	LFC+Morgan	0.59 $\pm$ 0.04	+0.04 [-0.08, 0.17]	0.282	+0.04 [-0.08, 0.17]	0.282	-0.00 [-0.10, 0.10]	0.998
SciPlex3	chemCPA	Post+Morgan	0.55 $\pm$ 0.07	+0.01 [-0.13, 0.14]	0.854	+0.01 [-0.13, 0.14]	0.854	+0.01 [-0.13, 0.14]	0.854
SciPlex3	scFoundation	LFC+Morgan	0.47 $\pm$ 0.11	-0.07 [-0.31, 0.17]	0.329	-0.07 [-0.31, 0.17]	0.329	-0.11 [-0.38, 0.15]	0.206
SciPlex3	scFoundation	Post+Morgan	0.69 $\pm$ 0.04	+0.14 [0.05, 0.24]	0.024*	+0.14 [0.05, 0.24]	0.024*	+0.14 [0.05, 0.24]	0.024*

Table 26: Within cell line (SciPlex3): Spearman ρ (mean $\pm$ s.d.; paired Δ [95% CI]. Primary: CPA vs Pre+Morgan / Average effect (Pearson) reports BH-FDR q within stratum; all other p are uncorrected exploratory. Sig: ** < 0.01 , * < 0.05).

Dataset	Model	Modality	Mean $\pm$ s.d.	Δ vs Pre [95% CI]	p/q	Δ vs Avg [95% CI]	p/q	Δ vs Noeff [95% CI]	p/q
SciPlex3	Average effect	LFC+Morgan	0.51 $\pm$ 0.00	-0.00 [-0.00, 0.00]	0.38	—	—	-0.03 [-0.08, 0.02]	0.116
SciPlex3	Average effect	Post+Morgan	0.51 $\pm$ 0.00	-0.00 [-0.00, 0.00]	0.237	—	—	-0.00 [-0.00, 0.00]	0.237
SciPlex3	CPA	LFC+Morgan	0.61 $\pm$ 0.09	+0.10 [-0.11, 0.31]	0.17	+0.10 [-0.11, 0.31]	0.17	+0.07 [-0.17, 0.31]	0.34
SciPlex3	CPA	Post+Morgan	0.59 $\pm$ 0.08	+0.09 [-0.12, 0.30]	0.219	+0.09 [-0.12, 0.30]	0.218	+0.09 [-0.12, 0.30]	0.219
SciPlex3	CPA End-to-End	-	0.26 $\pm$ 0.16	-0.25 [-0.64, 0.15]	0.118	-0.25 [-0.64, 0.15]	0.118	-0.28 [-0.67, 0.12]	0.0946
SciPlex3	ElasticNet	LFC+Morgan	0.67 $\pm$ 0.08	+0.17 [-0.04, 0.37]	0.0748	+0.17 [-0.04, 0.37]	0.0748	+0.13 [-0.12, 0.39]	0.154
SciPlex3	ElasticNet	Post+Morgan	0.64 $\pm$ 0.09	+0.14 [-0.09, 0.36]	0.121	+0.14 [-0.09, 0.36]	0.12	+0.14 [-0.09, 0.36]	0.121
SciPlex3	ElasticNet	Pre+Morgan	0.51 $\pm$ 0.00	—	—	—	—	—	—
SciPlex3	GEARS	LFC+Morgan	0.57 $\pm$ 0.04	+0.06 [-0.03, 0.16]	0.102	+0.06 [-0.03, 0.16]	0.102	+0.03 [-0.02, 0.08]	0.117
SciPlex3	GEARS	Post+Morgan	0.55 $\pm$ 0.02	+0.05 [0.00, 0.09]	0.0454*	+0.05 [0.00, 0.09]	0.0457*	+0.05 [0.00, 0.09]	0.0454*
SciPlex3	No effect	LFC+Morgan	0.54 $\pm$ 0.02	+0.03 [-0.02, 0.08]	0.116	+0.03 [-0.02, 0.08]	0.116	—	—
SciPlex3	No effect	Post+Morgan	0.51 $\pm$ 0.00	+0.00 [0.00, 0.00]	—	+0.00 [-0.00, 0.00]	0.237	—	—
SciPlex3	PRnet	LFC+Morgan	0.53 $\pm$ 0.08	+0.02 [-0.18, 0.22]	0.696	+0.02 [-0.18, 0.22]	0.696	-0.01 [-0.25, 0.22]	0.859
SciPlex3	PRnet	Post+Morgan	0.53 $\pm$ 0.07	+0.02 [-0.15, 0.19]	0.622	+0.02 [-0.15, 0.19]	0.62	+0.02 [-0.15, 0.19]	0.622
SciPlex3	chemCPA	LFC+Morgan	0.54 $\pm$ 0.04	+0.04 [-0.06, 0.13]	0.238	+0.04 [-0.06, 0.13]	0.238	+0.00 [-0.11, 0.12]	0.879
SciPlex3	chemCPA	Post+Morgan	0.50 $\pm$ 0.01	-0.01 [-0.04, 0.02]	0.443	-0.01 [-0.04, 0.02]	0.452	-0.01 [-0.04, 0.02]	0.443
SciPlex3	scFoundation	LFC+Morgan	0.43 $\pm$ 0.08	-0.08 [-0.28, 0.11]	0.212	-0.08 [-0.28, 0.11]	0.213	-0.11 [-0.34, 0.11]	0.162
SciPlex3	scFoundation	Post+Morgan	0.57 $\pm$ 0.10	+0.06 [-0.18, 0.30]	0.378	+0.06 [-0.18, 0.30]	0.378	+0.06 [-0.18, 0.30]	0.378

Table 27: Within cell line (SciPlex3): RMSE (mean $\pm$ s.d.; paired Δ [95% CI]. Primary: CPA vs Pre+Morgan / Average effect (Pearson) reports BH-FDR q within stratum; all other p are uncorrected exploratory. Sig: ** < 0.01 , * < 0.05).

Dataset	Model	Modality	Mean $\pm$ s.d.	Δ vs Pre [95% CI]	p/q	Δ vs Avg [95% CI]	p/q	Δ vs Noeff [95% CI]	p/q
SciPlex3	Average effect	LFC+Morgan	0.188 $\pm$ 0.044	-0.000 [-0.000, 0.000] +0.000	0.607	—	—	+0.000 [-0.009, 0.010] +0.000	0.864
SciPlex3	Average effect	Post+Morgan	0.188 $\pm$ 0.044	-0.000, 0.000 -0.032	0.687	—	—	[-0.000, 0.000] -0.031	0.687
SciPlex3	CPA	LFC+Morgan	0.156 $\pm$ 0.023	[-0.086, 0.023] -0.025	0.128	[-0.086, 0.023] -0.025	0.128	[-0.095, 0.033] -0.025	0.169
SciPlex3	CPA	Post+Morgan	0.163 $\pm$ 0.017	[-0.093, 0.044] +0.040	0.258	[-0.093, 0.044] +0.040	0.258	[-0.093, 0.044] +0.041	0.258
SciPlex3	CPA End-to-End	-	0.228 $\pm$ 0.052	[-0.000, 0.081] -0.050	0.0508	[-0.000, 0.081] -0.050	0.0508	[0.001, 0.080] -0.050	0.0471 *
SciPlex3	ElasticNet	LFC+Morgan	0.138 $\pm$ 0.013	[-0.131, 0.031] -0.039	0.117	[-0.131, 0.031] -0.039	0.117	[-0.140, 0.041] -0.039	0.142
SciPlex3	ElasticNet	Post+Morgan	0.149 $\pm$ 0.019	[-0.100, 0.023]	0.114	[-0.100, 0.023]	0.114	[-0.100, 0.023]	0.114
SciPlex3	ElasticNet	Pre+Morgan	0.188 $\pm$ 0.044	—	—	—	—	—	—
SciPlex3	GEARS	LFC+Morgan	0.171 $\pm$ 0.033	-0.017 [-0.075, 0.040] -0.010	0.324	-0.017 [-0.075, 0.040] -0.010	0.324	-0.017 [-0.082, 0.048] -0.010	0.379
SciPlex3	GEARS	Post+Morgan	0.178 $\pm$ 0.025	[-0.058, 0.037] -0.000	0.448	[-0.058, 0.037] -0.000	0.448	[-0.058, 0.037]	0.448
SciPlex3	No effect	LFC+Morgan	0.188 $\pm$ 0.048	[-0.010, 0.009] +0.000	0.863	[-0.010, 0.009] -0.000	0.864	—	—
SciPlex3	No effect	Post+Morgan	0.188 $\pm$ 0.044	[0.000, 0.000] -0.007	—	[-0.000, 0.000] -0.007	0.687	—	—
SciPlex3	PRnet	LFC+Morgan	0.181 $\pm$ 0.027	[-0.053, 0.039] -0.010	0.562	[-0.053, 0.039] -0.010	0.562	[-0.062, 0.048] -0.010	0.644
SciPlex3	PRnet	Post+Morgan	0.178 $\pm$ 0.023	[-0.066, 0.045] -0.008	0.516	[-0.066, 0.045] -0.008	0.516	[-0.066, 0.045] -0.008	0.516
SciPlex3	chemCPA	LFC+Morgan	0.180 $\pm$ 0.042	[-0.023, 0.006] -0.001	0.13	[-0.023, 0.006] -0.001	0.13	[-0.029, 0.013] -0.001	0.251
SciPlex3	chemCPA	Post+Morgan	0.187 $\pm$ 0.037	[-0.032, 0.031] +0.013	0.931	[-0.032, 0.031] +0.013	0.931	[-0.032, 0.031] +0.014	0.931
SciPlex3	scFoundation	LFC+Morgan	0.201 $\pm$ 0.025	[-0.033, 0.060] -0.024	0.346	[-0.033, 0.060] -0.024	0.346	[-0.042, 0.070] -0.024	0.404
SciPlex3	scFoundation	Post+Morgan	0.164 $\pm$ 0.025	[-0.085, 0.037]	0.235	[-0.085, 0.037]	0.235	[-0.085, 0.037]	0.235

McFarland per tissue

Table 28: Within tissue (McFarland): Pearson r (mean $\pm$ s.d.; paired Δ [95% CI]. Primary: CPA vs Pre+Morgan / Average effect (Pearson) reports BH-FDR q within stratum; all other p are uncorrected exploratory. Sig: ** < 0.01 , * < 0.05).

Dataset	Model	Modality	Mean $\pm$ s.d.	Δ vs Pre [95% CI]	p/q	Δ vs Avg [95% CI]	p/q	Δ vs Noeff [95% CI]	p/q
McFarland	Average effect	LFC+Morgan	0.67 $\pm$ 0.16	-0.01 [-0.04, 0.01]	0.143	—	—	-0.02 [-0.04, 0.00]	0.0769
McFarland	Average effect	Post+Morgan	0.67 $\pm$ 0.15	-0.01 [-0.03, 0.01]	0.373	—	—	-0.01 [-0.03, 0.01]	0.372
McFarland	CPA	LFC+Morgan	0.71 $\pm$ 0.14	+0.03 [0.00, 0.06]	0.0427*	+0.04 [0.02, 0.06]	0.00159**	+0.03 [-0.00, 0.05]	0.0664
McFarland	CPA	Post+Morgan	0.69 $\pm$ 0.14	+0.02 [-0.01, 0.04]	0.197	+0.03 [0.01, 0.05]	0.0248*	+0.02 [-0.01, 0.04]	0.197
McFarland	CPA End-to-End	-	0.52 $\pm$ 0.29	-0.13 [-0.24, -0.02]	0.0215*	-0.16 [-0.30, -0.02]	0.025*	-0.14 [-0.24, -0.04]	0.0104*
McFarland	ElasticNet	LFC+Morgan	0.73 $\pm$ 0.10	+0.08 [0.04, 0.13]	8.4e-04**	+0.11 [0.06, 0.17]	4.9e-04**	+0.08 [0.03, 0.12]	0.00315**
McFarland	ElasticNet	Post+Morgan	0.68 $\pm$ 0.13	+0.04 [0.00, 0.07]	0.0361*	+0.05 [-0.00, 0.10]	0.0548	+0.04 [0.00, 0.07]	0.0362*
McFarland	ElasticNet	Pre+Morgan	0.65 $\pm$ 0.14	—	—	—	—	—	—
McFarland	GEARS	LFC+Morgan	0.70 $\pm$ 0.12	+0.01 [-0.04, 0.07]	0.557	+0.03 [-0.02, 0.07]	0.264	+0.01 [-0.05, 0.06]	0.782
McFarland	GEARS	Post+Morgan	0.68 $\pm$ 0.13	-0.01 [-0.07, 0.06]	0.838	+0.00 [-0.05, 0.06]	0.874	-0.01 [-0.07, 0.06]	0.837
McFarland	No effect	LFC+Morgan	0.65 $\pm$ 0.15	+0.01 [-0.01, 0.02]	0.275	+0.02 [-0.00, 0.04]	0.0769	—	—
McFarland	No effect	Post+Morgan	0.65 $\pm$ 0.14	+0.00 [-0.00, 0.00]	0.484	+0.01 [-0.01, 0.03]	0.372	—	—
McFarland	PRnet	LFC+Morgan	0.67 $\pm$ 0.17	-0.02 [-0.04, -0.00]	0.0392*	+0.00 [-0.01, 0.01]	0.848	-0.02 [-0.05, 0.00]	0.0752
McFarland	PRnet	Post+Morgan	0.53 $\pm$ 0.23	-0.17 [-0.31, -0.04]	0.0167*	-0.14 [-0.23, -0.04]	0.00825**	-0.17 [-0.31, -0.04]	0.0167*
McFarland	chemCPA	LFC+Morgan	0.67 $\pm$ 0.16	-0.02 [-0.04, 0.01]	0.189	+0.00 [-0.01, 0.02]	0.558	-0.02 [-0.05, 0.01]	0.124
McFarland	chemCPA	Post+Morgan	0.68 $\pm$ 0.16	-0.01 [-0.03, 0.01]	0.199	+0.01 [-0.01, 0.03]	0.21	-0.01 [-0.03, 0.01]	0.199
McFarland	scFoundation	LFC+Morgan	0.71 $\pm$ 0.12	+0.02 [-0.02, 0.07]	0.237	+0.03 [-0.01, 0.08]	0.116	+0.02 [-0.03, 0.06]	0.399
McFarland	scFoundation	Post+Morgan	0.68 $\pm$ 0.12	-0.00 [-0.06, 0.06]	0.975	+0.01 [-0.04, 0.06]	0.768	-0.00 [-0.06, 0.06]	0.975

Table 29: Within tissue (McFarland): Spearman ρ (mean $\pm$ s.d.; paired Δ [95% CI]. Primary: CPA vs Pre+Morgan / Average effect (Pearson) reports BH-FDR q within stratum; all other p are uncorrected exploratory. Sig: ** < 0.01 , * < 0.05).

Dataset	Model	Modality	Mean $\pm$ s.d.	Δ vs Pre [95% CI]	p/q	Δ vs Avg [95% CI]	p/q	Δ vs Noeff [95% CI]	p/q
McFarland	Average effect	LFC+Morgan	0.54 $\pm$ 0.15	+0.01 [-0.03, 0.04]	0.661	—	—	+0.03 [-0.03, 0.08]	0.272
McFarland	Average effect	Post+Morgan	0.54 $\pm$ 0.16	+0.00 [-0.03, 0.03]	0.865	—	—	+0.06 [-0.03, 0.03]	0.864
McFarland	CPA	LFC+Morgan	0.58 $\pm$ 0.16	+0.03 [-0.01, 0.08]	0.118	+0.04 [0.01, 0.08]	0.018*	+0.03 [-0.01, 0.13]	0.108
McFarland	CPA	Post+Morgan	0.59 $\pm$ 0.16	+0.03 [-0.03, 0.08]	0.327	+0.05 [-0.01, 0.12]	0.11	+0.03 [-0.03, 0.08]	0.326
McFarland	CPA End-to-End	-	0.44 $\pm$ 0.25	-0.10 [-0.19, -0.02]	0.0199*	-0.13 [-0.25, -0.02]	0.0246*	-0.10 [-0.17, -0.02]	0.0121*
McFarland	ElasticNet	LFC+Morgan	0.63 $\pm$ 0.13	+0.09 [0.04, 0.13]	0.00107**	+0.09 [0.04, 0.13]	0.00138**	+0.09 [0.05, 0.13]	2.5e-04**
McFarland	ElasticNet	Post+Morgan	0.56 $\pm$ 0.20	+0.02 [-0.06, 0.09]	0.625	+0.01 [-0.13, 0.14]	0.909	+0.02 [-0.06, 0.09]	0.625
McFarland	ElasticNet	Pre+Morgan	0.54 $\pm$ 0.15	—	—	—	—	—	—
McFarland	GEARS	LFC+Morgan	0.58 $\pm$ 0.19	+0.00 [-0.11, 0.11]	0.962	-0.01 [-0.11, 0.09]	0.868	-0.01 [-0.12, 0.11]	0.915
McFarland	GEARS	Post+Morgan	0.56 $\pm$ 0.19	-0.02 [-0.12, 0.08]	0.728	+0.00 [-0.08, 0.08]	0.945	-0.02 [-0.12, 0.08]	0.729
McFarland	No effect	LFC+Morgan	0.54 $\pm$ 0.19	-0.00 [-0.06, 0.05]	0.91	-0.03 [-0.08, 0.03]	0.272	—	—
McFarland	No effect	Post+Morgan	0.54 $\pm$ 0.15	-0.00 [-0.00, 0.00]	0.33	-0.00 [-0.03, 0.03]	0.864	—	—
McFarland	PRnet	LFC+Morgan	0.54 $\pm$ 0.15	-0.00 [-0.03, 0.02]	0.735	+0.00 [-0.02, 0.02]	0.795	+0.02 [-0.05, 0.08]	0.571
McFarland	PRnet	Post+Morgan	0.55 $\pm$ 0.19	-0.04 [-0.12, 0.03]	0.247	+0.01 [-0.08, 0.10]	0.772	-0.04 [-0.12, 0.03]	0.248
McFarland	chemCPA	LFC+Morgan	0.54 $\pm$ 0.15	-0.01 [-0.06, 0.05]	0.732	-0.00 [-0.03, 0.02]	0.812	+0.01 [-0.03, 0.05]	0.498
McFarland	chemCPA	Post+Morgan	0.57 $\pm$ 0.22	-0.02 [-0.08, 0.03]	0.338	+0.04 [-0.05, 0.12]	0.389	-0.02 [-0.08, 0.03]	0.338
McFarland	scFoundation	LFC+Morgan	0.58 $\pm$ 0.17	-0.01 [-0.09, 0.08]	0.894	-0.01 [-0.09, 0.07]	0.855	-0.01 [-0.11, 0.08]	0.749
McFarland	scFoundation	Post+Morgan	0.55 $\pm$ 0.18	-0.03 [-0.13, 0.07]	0.52	-0.01 [-0.08, 0.07]	0.874	-0.03 [-0.13, 0.07]	0.52

Table 30: Within tissue (McFarland): RMSE (mean $\pm$ s.d.; paired Δ [95% CI]. Primary: CPA vs Pre+Morgan / Average effect (Pearson) reports BH-FDR q within stratum; all other p are uncorrected exploratory. Sig: ** < 0.01 , * < 0.05).

Dataset	Model	Modality	Mean $\pm$ s.d.	Δ vs Pre [95% CI]	p/q	Δ vs Avg [95% CI]	p/q	Δ vs Noeff [95% CI]	p/q
McFarland	Average effect	LFC+Morgan	0.143 $\pm$ 0.048	+0.003 [-0.002, 0.008]	0.207	—	—	+0.003 [-0.003, 0.009]	0.336
McFarland	Average effect	Post+Morgan	0.143 $\pm$ 0.049	+0.003 [-0.003, 0.008]	0.346	—	—	+0.003 [-0.003, 0.008]	0.346
McFarland	CPA	LFC+Morgan	0.135 $\pm$ 0.046	-0.005 [-0.010, 0.001]	0.117	-0.008 [-0.011, -0.005]	1.9e-05**	-0.005 [-0.012, 0.002]	0.171
McFarland	CPA	Post+Morgan	0.140 $\pm$ 0.044	-0.002 [-0.007, 0.003]	0.35	-0.003 [-0.006, 0.001]	0.153	-0.002 [-0.007, 0.003]	0.351
McFarland	CPA End-to-End	-	0.178 $\pm$ 0.062	+0.026 [0.007, 0.044]	0.00988**	+0.031 [0.006, 0.056]	0.0183*	+0.025 [0.008, 0.043]	0.00658**
McFarland	ElasticNet	LFC+Morgan	0.136 $\pm$ 0.030	-0.016 [-0.023, -0.010]	2.8e-05**	-0.022 [-0.030, -0.014]	4.9e-05**	-0.016 [-0.023, -0.010]	7.8e-05**
McFarland	ElasticNet	Post+Morgan	0.145 $\pm$ 0.033	-0.007 [-0.012, -0.003]	0.00397**	-0.010 [-0.018, -0.003]	0.00926**	-0.007 [-0.012, -0.003]	0.00398**
McFarland	ElasticNet	Pre+Morgan	0.153 $\pm$ 0.037	—	—	—	—	—	—
McFarland	GEARS	LFC+Morgan	0.160 $\pm$ 0.041	+0.016 [0.002, 0.030]	0.0265*	+0.010 [-0.004, 0.025]	0.149	+0.017 [0.002, 0.032]	0.0323*
McFarland	GEARS	Post+Morgan	0.158 $\pm$ 0.040	+0.014 [0.003, 0.026]	0.0202*	+0.008 [-0.005, 0.021]	0.233	+0.014 [0.003, 0.026]	0.0202*
McFarland	No effect	LFC+Morgan	0.153 $\pm$ 0.039	-0.000 [-0.002, 0.002]	0.964	-0.003 [-0.009, 0.003]	0.336	—	—
McFarland	No effect	Post+Morgan	0.153 $\pm$ 0.037	-0.000 [-0.000, 0.000]	0.728	-0.000 [-0.008, 0.003]	0.346	—	—
McFarland	PRnet	LFC+Morgan	0.143 $\pm$ 0.052	+0.004 [-0.000, 0.009]	0.0707	-0.000 [-0.004, 0.003]	0.84	+0.004 [-0.002, 0.010]	0.161
McFarland	PRnet	Post+Morgan	0.197 $\pm$ 0.100	+0.066 [0.008, 0.124]	0.0297*	+0.054 [0.014, 0.095]	0.0106*	+0.066 [0.008, 0.124]	0.0297*
McFarland	chemCPA	LFC+Morgan	0.143 $\pm$ 0.049	+0.004 [-0.002, 0.009]	0.185	+0.000 [-0.002, 0.002]	0.98	+0.004 [-0.003, 0.010]	0.278
McFarland	chemCPA	Post+Morgan	0.143 $\pm$ 0.051	+0.005 [-0.002, 0.009]	0.175	+0.002 [-0.002, 0.003]	0.693	+0.006 [-0.002, 0.009]	0.175
McFarland	scFoundation	LFC+Morgan	0.151 $\pm$ 0.036	+0.010 [-0.005, 0.016]	0.287	+0.004 [-0.010, 0.013]	0.767	+0.010 [-0.006, 0.018]	0.286
McFarland	scFoundation	Post+Morgan	0.154 $\pm$ 0.038	[-0.002, 0.023]	0.103	[-0.008, 0.016]	0.479	[-0.002, 0.023]	0.103